

Jio Subscriber Churn Intelligence

Workstream 1 — Exploratory Data Analysis & Business Diagnostics Executive Brief

Executive Summary

This brief summarizes findings from a comprehensive exploratory analysis of 64,738 Jio subscribers. The current 30-day churn rate stands at **1.73%**, reflecting a highly imbalanced but business-critical retention challenge. The analysis identifies the strongest churn drivers across network experience, pricing, recharge behavior, and customer engagement, providing a factual foundation for the predictive modelling work planned in Workstream 2.

Dataset Overview

Metric	Value
Total Subscribers	64,738
Total Data Columns	43
Average ARPU (Last Month)	INR 213.70
Baseline 30-Day Churn Rate	1.73%
Churned Subscribers	1,122
Retained Subscribers	63,616

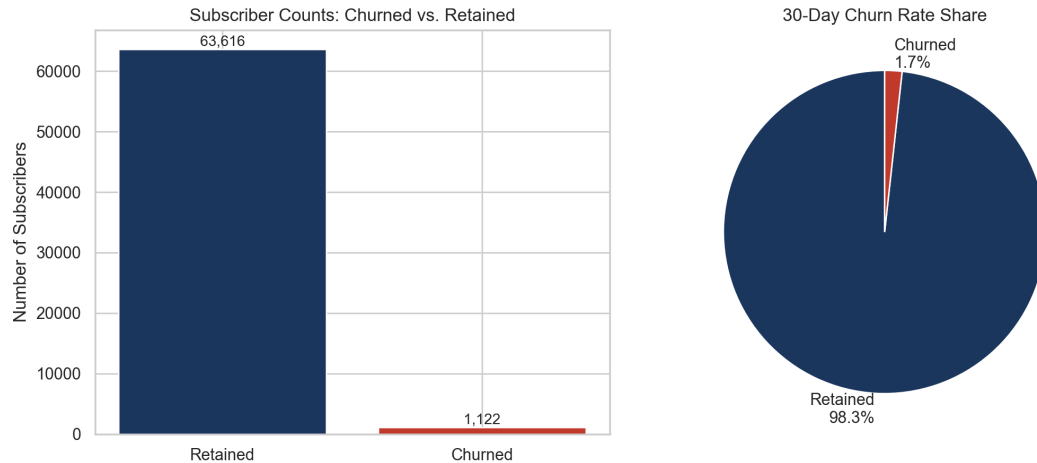
Key Takeaways

- Dataset covers 64,738 subscribers with a baseline 30-day churn rate of 1.73% (1,122 churned vs. 63,616 retained), an imbalance of roughly 1 churner per 57 retained subscribers.
- 'Uttar Pradesh East' circle shows the highest churn concentration at 2.8%, making it the top-priority region for retention campaigns.
- Churn risk accelerates sharply once the recharge gap reaches the '90+'-day band, marking the 'silent churn' threshold for proactive outreach.
- The '<150' INR ARPU bracket has the highest churn rate at 3.2%, pointing to pricing/value-perception risk in that tier.
- 'days_since_last_recharge' is the numeric feature most correlated with churn ($r = 0.22$), flagging it as a strong candidate signal for the predictive model.

Target Variable Analysis & Churn Drivers

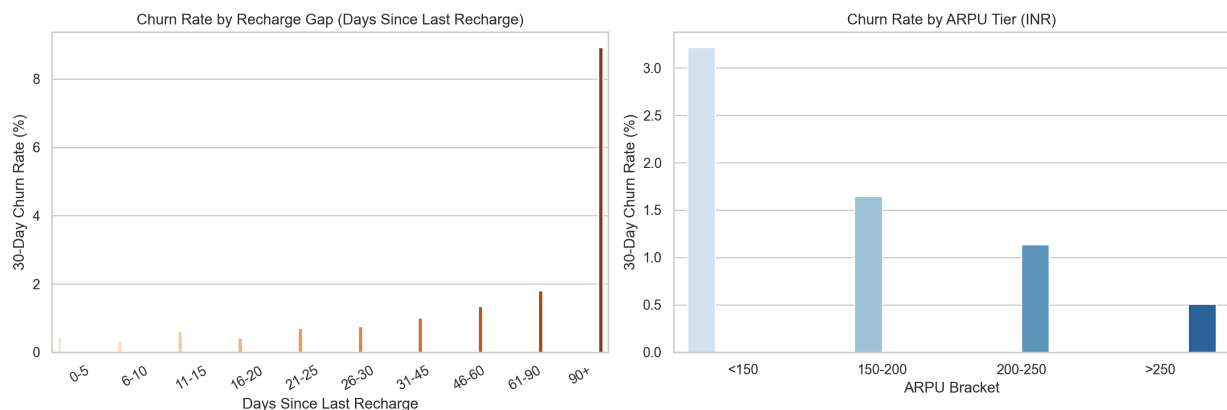
Only **1.73%** of subscribers churned in the last 30 days, but the business impact is concentrated: certain recharge behaviors and pricing tiers show materially higher risk. The charts below highlight where retention effort should be focused first.

Target Variable Distribution: churn_flag_30d



Recharge Gaps & ARPU Tiers as Leading Risk Signals

Key Churn Risk Factors



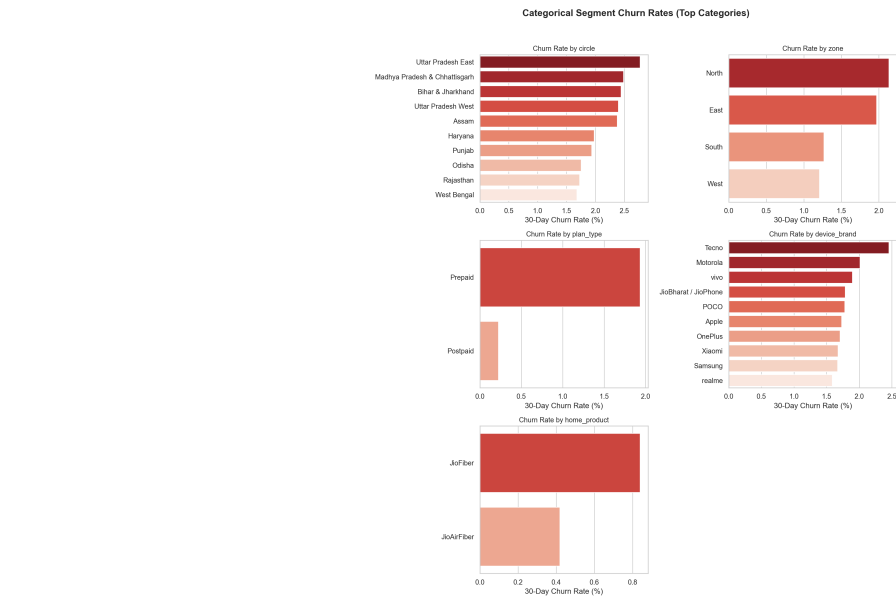
Subscribers who go significantly longer without recharging show a sharply higher chance of churning in the following 30 days — this 'silent churn' window is the clearest early-warning signal available today. Similarly, churn is not spread evenly across ARPU (average revenue per user) tiers: certain price brackets show elevated risk, suggesting a value-for-money gap that targeted offers can address.

Variable-by-Variable Deep Dive

The table below translates the most predictive raw data fields into plain-English business meaning, alongside their typical (average) values across the subscriber base.

Variable	Business Meaning	Average	Median
arpu_last_month_inr	Average revenue per user last month - the core measure of subscriber value.	213.7	188.4
tenure_months	How long a subscriber has stayed with Jio - early-tenure customers are typically more churn-prone.	32.4	27.0
days_since_last_recharge	Days since the last recharge - a direct proxy for engagement and 'silent churn' risk.	40.2	25.0
avg_recharge_gap_days	Typical time between recharges - larger gaps signal weakening commitment.	70.1	36.7
data_gb_last_month	Data consumption last month - a proxy for how 'sticky' the connection is for daily use.	40.7	31.6
complaints_6m	Number of complaints in the last 6 months - a direct dissatisfaction signal.	0.6	0.0
payment_failures_6m	Failed payment attempts - often precede voluntary or involuntary churn.	0.4	0.0
outgoing_to_competitor_pct	Share of usage directed to competitor networks - an early switching signal.	30.2	26.4

Which Customer Segments Carry the Most Risk?



Churn is not uniform across geography, plan type, or device brand. The chart above ranks the segments with the highest 30-day churn rates, helping prioritize which customer groups need retention attention first.

Strategic Recommendations & Next Steps

Immediate Retention Actions

- Launch proactive, automated outreach (SMS/app nudge/callback offer) to subscribers approaching the identified 'silent churn' recharge-gap threshold, before they lapse.
- Review pricing and value bundles for the ARPU tier(s) showing the highest churn rate, e.g. via loyalty discounts, data top-ups, or plan right-sizing.
- Prioritize retention campaigns in the highest-churn circles/zones and device segments identified in this analysis, rather than a one-size-fits-all approach.
- Fast-track resolution of unresolved complaints — unresolved issues are a strong, actionable churn signal that can be fixed operationally.
- Monitor payment-failure patterns and offer flexible retry/payment options to reduce involuntary churn.

Next Steps: Workstream 2 — Predictive Modelling

- Build a supervised churn-propensity model (e.g. gradient boosting or logistic regression baseline) using the engineered risk signals surfaced in this EDA (recharge gap, ARPU tier, complaints, network KPIs). With an observed imbalance of roughly 57:1 (retained:churned), class-imbalance-aware techniques (e.g. class weighting, SMOTE, or threshold tuning) will be essential.
- Engineer time-windowed behavioral features (trend in ARPU, recharge cadence changes, recent complaint velocity) rather than relying on single snapshot values.
- Validate the model with business-relevant metrics (precision at top-risk decile, recall, lift) rather than raw accuracy, given the rarity of churn events.
- Establish a monthly scoring pipeline that feeds a prioritized, actionable retention worklist to the customer-care and marketing teams.
- Set up a feedback loop to track intervention outcomes (offer redemption, post-intervention churn) to continuously improve targeting precision.